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FIRST Function

Rules of FIRST:

1. If x is a terminal, then $\text{FIRST}(x) = \{ 'x' \}$
2. If $x \rightarrow \epsilon$, is a production rule, then add ϵ to $\text{FIRST}(x)$.
3. If $X \rightarrow Y_1 Y_2 Y_3 \dots Y_n$ is a production,
 - a. $\text{FIRST}(X) = \text{FIRST}(Y_1)$
 - b. If $\text{FIRST}(Y_1)$ contains ϵ then $\text{FIRST}(X) = \{ \text{FIRST}(Y_1) - \epsilon \} \cup \{ \text{FIRST}(Y_2) \}$
 - c. If $\text{FIRST}(Y_i)$ contains ϵ for all $i = 1$ to n , then add ϵ to $\text{FIRST}(X)$.

Terminal & Non-terminal

- All the capital letters are non-terminal (A,B,C,D etc).
- Except capital letters, all other characters and operators are terminal (*,+,-,(,),a,b,c,d etc).

Rules for FIRST

$$T \rightarrow t$$

For the above production rule, $\text{FIRST}(T) \rightarrow \{t\}$

$$T \rightarrow tB \mid \epsilon$$

For the above production rule, $\text{FIRST}(T) \rightarrow \{t, \epsilon\}$

$$T \rightarrow At$$

$$A \rightarrow aT$$

For the above production rules, $\text{FIRST}(T) \rightarrow \text{FIRST}(A) \rightarrow \{a\}$

$$T \rightarrow At$$

$$A \rightarrow aT \mid \epsilon$$

For the above production rules, $\text{FIRST}(T) \rightarrow \text{FIRST}(A) \rightarrow \{a, \epsilon\}$

But A replaced by ϵ so $\text{FIRST}(T) \rightarrow \{a, t\}$

Example:

Production rules are given: $E \rightarrow TE'$

$$E' \rightarrow +T E' \mid \epsilon$$
$$T \rightarrow F T'$$
$$T' \rightarrow *F T' \mid \epsilon$$
$$F \rightarrow \text{id} \mid (E)$$

Now find sets of FIRST Function

FIRST:

Production	FIRST
$E \rightarrow TE'$	$\{ \text{id}, (\}$
$E' \rightarrow +T E' \mid \epsilon$	$\{ +, \epsilon \}$
$T \rightarrow F T'$	$\{ \text{id}, (\}$
$T' \rightarrow *F T' \mid \epsilon$	$\{ *, \epsilon \}$
$F \rightarrow \text{id} \mid (E)$	$\{ \text{id}, (\}$

$\text{FIRST}(E) \rightarrow \text{FIRST}(T) \rightarrow \text{FIRST}(F) \rightarrow \{ \text{id}, (\}$

More Exercises:

Production rules are given: $S \rightarrow Bb \mid cd$

$B \rightarrow aB \mid \epsilon$

$C \rightarrow cC \mid \epsilon$

Now find sets of FIRST

More Exercises:

Production rules are given: $S \rightarrow ABCDE$

$A \rightarrow a \mid \epsilon$

$B \rightarrow b \mid \epsilon$

$C \rightarrow c$

$D \rightarrow d \mid \epsilon$

$E \rightarrow e \mid \epsilon$

Now find sets of FIRST

Thank You